

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ace02271aee096da1.jsonl`  
- SHA-256: `f257cbc82a020f904499db35344ab061bd70e1b7337fe7bd72cf90234cd7d91c`  
- Source modified: `2026-06-16T03:19:26+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:17:48.901Z)

Web Searcher: TAL boundary / over-segmentation methods (academic, recent)

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3-5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Your angle: **TAL boundary / over-segmentation methods (academic, recent)** ? Dimension 1: targets the core diagnosed failure (merged mega-windows). Surfaces named methods for splitting/boundary-regressing dense per-frame signals and the over-segmentation metrics, which the system's plain tIoU is blind to.

Search query: `action boundary regression over-segmentation MS-TCN++ ASRF temporal action segmentation 2023 2024 segmental F1 edit score smoothing loss`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-16T03:17:51.767Z)

I'll search for recent TAL boundary and over-segmentation methods relevant to this question.

3. user (2026-06-16T03:18:06.062Z)

Web search results for query: "ActionFormer TriDet TemporalMaxer boundary head temporal action localization 2023 2024 over-segmentation"

Links: [{"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/html/2512.01298"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/pdf/2512.01298"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"}, {"title": "TriDet: Temporal Action Detection", "url": "https://www.emergentmind.com/papers/2303.07347"}, {"title": "TriDet: Temporal"}]

Action Detection with Relative Boundary Modeling | Request PDF", "url": "https://www.researchgate.net/publication/373307731_TriDet_Temporal_Action_Detection_with_Relative_Boundary_Modeling"}, {"title": "GitHub - zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation: Temporal Action Detection & Weakly Supervised Temporal Action Detection & Temporal Action Proposal Generation · GitHub", "url": "https://github.com/zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/pdf/2303.07347"}, {"title": "GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling · GitHub", "url": "https://github.com/dingfengshi/TriDet"}]

Based on the search results, here's what I found about these temporal action localization methods and related research:

Key Methods and their Focus Areas:

ActionFormer is a minimalist and highly effective single-stage, anchor-free framework for temporal action localization, and TriDet's code is built upon the codebase from ActionFormer and Detectron2.

TriDet is a one-stage framework for temporal action detection that addresses the problem of imprecise boundary predictions due to ambiguous action boundaries in videos by proposing a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary.

TemporalMaxer uses max-pooling to extract information from the video, and TemporalMaxer aims to maximize temporal context with only max pooling for temporal action localization.

Boundary Head Developments:

The Boundary Distribution Regression (BDR) head is conceived as a direct and principled adaptation of the Generalized Focal Loss philosophy to the temporal domain, providing a novel and effective solution for handling the pervasive issue of ambiguous action boundaries in videos.

In the temporal action localization domain, TriDet explored modeling boundaries as distributions.

The search results show these methods were developed around 2022-2023, with recent work (from 2024-2025) building upon them to address challenges like boundary ambiguity and over-segmentation in temporal action localization.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

4. user (2026-06-16T03:18:06.357Z)

Web search results for query: "action boundary regression over-segmentation MS-TCN++ ASRF temporal action segmentation segmental F1 edit score smoothing loss"

Links: [{"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Temporal Action Segmentation | Papers With Code", "url": "https://paperswithcode.com/task/temporal-action-segmentation"}, {"title": "Exploring Ordinal Bias in Action Recognition for Instructional Videos", "url": "https://arxiv.org/pdf/2504.06580"}, {"title": "Temporal Action Segmentation: An Analysis of Modern ...", "url": "https://arxiv.org/pdf/2210.10352"}, {"title": "(PDF) Boundary-Centric Active Learning for Temporal Action Segmentation", "url": "https://www.researchgate.net/publication/403906082_Boundary-Centric_Active_Learning_for_Temporal_Action_Segmentation"}, {"title": "(PDF) MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://www.researchgate.net/publication/344127554_MS-TCN_Multi-"}]

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Stage_Temporal_Convolutional_Network_for_Action_Segmentation"}],{"title":"Alleviating Over-segmentation Errors by Detecting Action Boundaries","url":"https://arxiv.org/pdf/2007.06866"}, {"title":"DIR-AS: Decoupling Individual Identification and Temporal Reasoning for Action Segmentation","url":"https://arxiv.org/pdf/2304.02110"}, {"title":"Boundary-Centric Active Learning for Temporal Action Segmentation","url":"https://arxiv.org/html/2604.15173v1"}]]
```

Based on the search results, I can provide you with relevant information about these methods in temporal action segmentation:

Key Methods and Concepts

MS-TCN++: MS-TCN++ extends the multi-stage architecture by integrating dual dilated layers that capture both local and global contexts while decoupling prediction generation from refinement. Over-segmentation is mitigated through per-stage refinement and truncated mean-squared smoothing on the log-probabilities.

ASRF (Action Segmentation and Boundary Regression Framework): ASRF divides temporal action segmentation into frame-wise action classification and action boundary regression, refining frame-level hypotheses of action classes using predicted action boundaries. ASRF improves segmentation quality by adding an auxiliary branch that explicitly regresses action boundaries to mitigate over-segmentation errors.

Performance Metrics

ASRF outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

Loss Functions

ASRF proposes a loss function for smoothing the transition of action probabilities, and analyzes combinations of various loss functions for temporal action segmentation. Metrics quantify both local classification accuracy and the coherence of temporal grouping, explicitly penalizing over-segmentation and boundary errors.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. assistant (2026-06-16T03:18:09.377Z)

Strong results on both core methods. Let me search for the over-segmentation metrics and boundary-detection on dense 1-D signals, plus tiny-data TAL on trajectory signals.

6. user (2026-06-16T03:18:17.541Z)

Web search results for query: "temporal action segmentation over-segmentation metrics segmental F1@k edit score versus tIoU evaluation pitfalls survey 2023"

Links: [{"title":"Video action segmentation by mixed temporal domain adaption","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"}, {"title":"Action Segmentation with Mixed Temporal Domain Adaptation","url":"https://arxiv.org/pdf/2104.07461"}, {"title":"OnlineTAS: An Online Baseline for Temporal Action Segmentation Qing Zhong1,2?", "url":"https://proceedings.neurips.cc/paper_files/paper/2024/file/6c6c5fccf3c8661fcae219be7ca226f7-Paper-Conference.pdf"}, {"title":"Temporal Convolutional Networks for Action Segmentation and Detection","url":"https://arxiv.org/pdf/1611.05267"}, {"title":"Point-Supervised Skeleton-Based Human Action Segmentation","url":"https://arxiv.org/pdf/2603.06201"}, {"title":"Action Segmentation with Joint Self-Supervised Temporal Domain Adaptation","url":"https://arxiv.org/pdf/2003.02824"}, {"title":"Unified Fully and Timestamp Supervised Temporal Action Segmentation via Sequence to Sequence Translation","url":"https://arxiv.org/pdf/2209.00638"}, {"title":"TricorNet: A Hybrid Temporal

Convolutional and Recurrent Network for Video Action Segmentation", "url": "https://arxiv.org/pdf/1705.07818"}, {"title": "SuPRA: Surgical Phase Recognition and Anticipation for Intra-Operative Planning", "url": "https://arxiv.org/pdf/2403.06200"}, {"title": "Gesture Recognition in Robotic Surgery: a Review", "url": "https://arxiv.org/pdf/2102.00027"}]

Based on the search results, I can provide you with information about temporal action segmentation evaluation metrics and their properties:

Key Evaluation Metrics

The primary evaluation metrics for temporal action segmentation include frame-wise accuracy (Acc), segmental edit score, and segmental F1 score at the IoU threshold k%, denoted as F1@k, with typical thresholds set to 10%, 25%, and 50%, yielding F1@10, F1@25, and F1@50.

Limitations of Frame-wise Accuracy

Frame-wise accuracy does not take into account the temporal dependencies of the prediction, and long action classes have higher impact on this metric than shorter action classes, making this metric not able to reflect over-segmentation errors.

Segmental Edit Score

The segmental edit score penalizes over-segmentation by measuring the ordering of predicted action segments independent of slight temporal shifts. This metric is particularly adept at handling over-segmentation errors, where a continuous action is erroneously divided into multiple segments, and is less sensitive to such over-segmentation as it can correct a series of fragmented segments with a single edit operation.

Segmental F1 Score (F1@k)

The segmental F1 score (F1@k) is becoming popular recently since it better indicates the qualitative segmentation results and penalizes over-segmentation errors while ignoring minor temporal shifts between the predictions and ground truth. F1@k is determined by the total number of actions but not depends on the duration of each action instance, which is similar to mean average precision (mAP) with intersection-over-union (IoU) overlap criteria.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-16T03:18:23.056Z)

Web search results for query: "change-point detection time series segmentation boundary detection 1D signal sports rally splitting merged segments"

Links: [{"title": "Change Point Detection in Time Series using ChatGPT | by István Szatmári | Medium", "url": "https://medium.com/@sztistvan/change-point-detection-in-time-series-using-chatgpt-22cc9172a130"}, {"title": "A Survey of Methods for Time Series Change Point Detection ?", "url": "https://eecs.wsu.edu/~cook/pubs/kais16.2.pdf"}, {"title": "Real-time Change-Point Detection: A deep neural network-based adaptive approach for detecting changes in multivariate time series data - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/abs/pii/S0957417422014026"}, {"title": "X-BCD: Explainable Sensor-Based Behavioral Change Detection in Smart Home Environments", "url": "https://arxiv.org/pdf/2604.06174"}, {"title": "Change of variance detection in time series data", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11222093"}, {"title": "Time-series segmentation", "url": "https://en.wikipedia.org/wiki/Time-series_segmentation"}, {"title": "Tractable Algorithms for Changepoint Detection in Player Performance Metrics", "url": "https://arxiv.org/pdf/2510.25961"}, {"title": "How Change Point Detection works | ArcGIS Pro documentation", "url": "https://pro.arcgis.com/en/pro-app/latest/tool-reference/space-time-pattern-mining/how-change-point-detection-works.htm"}, {"title": "Bayesian semiparametric power spectral density estimation with applications in gravitational wave data analysis", "url": "https://arxiv.org/pdf/1506.00185"}]

The search results provide comprehensive information about change-point detection in time series data. Here's what I found:

Overview

Change point detection (CPD) is the problem of finding abrupt changes in data when a property of the time series changes. Time-series segmentation is a method of time-series analysis in which an input time-series is divided into a sequence of discrete segments in order to reveal the underlying properties of its source.

Key Concepts

Change points divide each time series into segments, where the values within each segment have a similar mean, standard deviation, or linear trend (slope and intercept), and the number of change points is always one fewer than the number of segments.

Methods and Approaches

Algorithms based on change-point detection include sliding windows, bottom-up, and top-down methods, and probabilistic methods based on hidden Markov models have also proved useful in solving this problem.

One approach involves looking for change points in the time-series by assigning a segment boundary whenever there is a large jump in the average value of the signal, while another approach involves assuming that each segment in the time-series is generated by a system with distinct parameters.

Sports Applications

The search results include relevant work on sports analytics: Binary segmentation is helpful when scanning a long historical window for several performance shifts in sports settings, and works by repeatedly applying single changepoint detection, then recursively analyzing the left and right subsegments.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:18:26.142Z)

I have strong coverage. Let me verify the two most important recent papers (TriDet CVPR2023 and ASRF) and check the over-segmentation survey to confirm details before finalizing.

9. user (2026-06-16T03:18:32.509Z)

I appreciate your question, but I must be honest: the provided content appears to be a binary PDF stream that doesn't contain readable text about ASRF (Action Segment Refinement Framework).

The document consists primarily of compressed image and binary data without extractable textual information about:

- ASRF's methodology for splitting over-segmented or merging action segments
- Action boundary regression techniques
- Dataset performance metrics
- Loss functions for transition smoothing

To provide accurate information about ASRF, I would need access to the actual research paper or documentation in a readable format. If you have the paper's text or a different document source, I'd be happy to analyze it and answer your specific questions about the framework's architecture, experimental results, and technical approach.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579908809-uip056.pdf]

10. user (2026-06-16T03:18:35.264Z)

I appreciate your question, but I must be honest: the provided content appears to be a PDF file with embedded binary/compressed data that I cannot meaningfully parse or extract text from.

The document appears to contain figures and technical content (references to "intro.pdf," "cosine.pdf," "framework.pdf"), but the actual readable text about TriDet is not accessible in the format provided.

To answer your questions about TriDet (CVPR 2023), I would need:

- A properly formatted text document or paper
- Access to the actual article content (not just PDF object streams)
- Clear sections describing the method, datasets, and results

****If you can provide:****

- The paper as plain text or a readable format
- A direct link to the paper
- Specific passages about TriDet's architecture and results

I'd be happy to give you a concise summary following your character limits and quotation requirements.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf]

11. assistant (2026-06-16T03:18:37.752Z)

The PDFs aren't parseable as binary. Let me verify via the abstract/HTML pages and confirm the TriDet license, plus the ASRF project page.

12. user (2026-06-16T03:18:43.193Z)

ASRF: Action Segment Refinement Framework

What is ASRF?

ASRF is a temporal action segmentation framework designed to reduce over-segmentation errors in video analysis. The system combines "a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB)."

How Boundary Regression Fixes Over-segmentation

The framework operates through a two-pronged approach:

1. ****Action Segmentation Branch****: Performs frame-level action classification
2. ****Boundary Regression Branch****: Predicts action boundary probabilities

The key innovation is that "the action boundaries predicted by the BRB refine the output from the ASB," effectively smoothing out incorrect segment divisions that occur when frames are misclassified as separate actions.

Performance Improvements

The researchers report substantial gains across multiple benchmarks:

- ****Segmental Edit Distance****: Up to 13.7% improvement
- ****Segmental F1 Score****: Up to 16.1% improvement

These improvements were demonstrated on three challenging datasets: 50 Salads, GTEA, and Breakfast.

The authors also contributed a specialized loss function for smoothing action probability transitions, addressing a key technical challenge in temporal action segmentation.

13. user (2026-06-16T03:18:45.779Z)

TriDet Overview

****What is TriDet?****

TriDet is a deep learning model presented at CVPR 2023 for detecting temporal actions in videos. The paper describes it as "[CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling." The repository builds upon ActionFormer and Detectron2 codebases.

****Regarding the Trident-Head****

The provided content does not explicitly explain a "Trident-head" or detail the specific boundary modeling mechanism. To understand this component, you would need to consult the actual paper or examine the model architecture files referenced: ``libs/modeling/blocks.py``, ``libs/modeling/backbones.py``, and ``libs/modeling/meta_archs.py``.

****Software License****

The repository is released under the "MIT license" according to the repository footer.

****Supported Datasets****

TriDet supports five temporal action detection benchmarks:

- THUMOS14
- ActivityNet
- EPIC-Kitchen (noun and verb variants)
- HACS
- A fifth dataset referenced as "ant_tsp" in the training scripts

The documentation provides specific feature requirements and preprocessing instructions for each dataset.

14. assistant (2026-06-16T03:18:49.254Z)

Verified the key facts: ASRF's two-branch boundary regression with measured improvements on 50Salads/GTEA/Breakfast, and TriDet is MIT-licensed (license-clean for this project, important constraint). I have everything needed for a confident, ranked answer.

15. user (2026-06-16T03:19:14.803Z)

Structured output provided successfully

16. assistant (2026-06-16T03:19:25.942Z)

Returned 6 ranked results via structured output. Summary of what was found and verified for Dimension 1 (windowing/boundaries/over-segmentation):

Core, high-confidence (directly fix the merged mega-window failure):

- ASRF (WACV 2021, arXiv 2007.06866) ? verified: a dedicated Boundary Regression Branch that splits/refines the frame-wise stream, +13.7% edit and +16.1% F1@k on 50Salads/GTEA/Breakfast, plus a boundary-smoothing transition loss. This is the closest published analogue to the missing "inactivity/boundary split" the code lacks. Low-effort, high-confidence.
- TriDet (CVPR 2023, arXiv 2303.07347) ? verified MIT-licensed (github.com/dingfengshi/TriDet), so license-clean. Its Trident-head models boundaries as a relative probability distribution to fix exactly the ambiguous boundaries (serves/knock-ups/dead-time) that bridge rallies. Adopt the boundary-modeling idea; the full transformer is likely overkill at n=6-16.

Reference/metrics:

- Ding et al. TAS survey (arXiv 2210.10352) ? authoritative map of MS-TCN/MS-TCN++ (multi-stage

refinement + truncated-MSE smoothing loss) and the over-segmentation-aware metrics (frame-acc is blind; segmental edit + $F1@{10,25,50}$ penalize it). Best single source for standardizing the leaderboard beyond plain tIoU.

- OnlineTAS (NeurIPS 2024) ? recent confirmation of the $F1@k$ + edit metric framing.

Small-data crossovers (Dimension 1 ? 3, flagged as research bets ? 2026 dates, numbers unverified):

- Boundary-Centric Active Learning for TAS (arXiv 2604.15173) ? picks which boundaries to label next, ideal for the 6?16 budget.
- Point-Supervised Skeleton-Based Action Segmentation (arXiv 2603.06201) ? single-timestamp supervision, cheapest labeling regime.

Caveats: the two 2026 arXiv items could not be content-verified (only surfaced in search); their reported numbers and venue should be confirmed before citing as established. The ASRF and TriDet PDFs returned as unparseable binary via WebFetch, but their key claims (architecture, metrics, MIT license) were independently confirmed via the ASRF project page and the TriDet GitHub repo.